

Litian Liang

RESEARCH INTERESTS

Large Language Model, Foundation Models, RNA Design, Protein Design

EDUCATION

Carnegie Mellon University

Master of Computational Biology

GPA: 4.12/4.33

Relevant Courses: Computational Methods for Biological Modeling and Simulation, String Algorithm, Computational Methods for Proteogenomics and Metabolomics, Introduction to Deep Learning, Computational Molecular Biology and Genomics

Pittsburgh, PA, USA

May 2025

Fudan University

Bachelor of Data Science

GPA: 3.57/4.0

2nd Scholarship for Outstanding Academic Performance

Relevant Courses: Computer Vision, Neural Network and Deep Learning, Principle of Computer Engineering, Biostatistics, Medical Image Processing, Numerical Algorithms with Case Studies

Shanghai, China

June 2023

RESEARCH EXPERIENCES

Carnegie Mellon University

Developing ARCADE with activation engineering for controllable codon design

- Contributed to proposing ARCADE, the first deep learning framework enabling controllable codon design through activation engineering without additional training
- Designed steering vectors that provide continuous control over codon adaptation index (CAI), mRNA stability, immune-related motifs, and protein expression

Pittsburgh, PA, USA

Feb 2025 – May 2025

Enhancing Genomic Language Models with Adaptive RainbowDQN for Optimal K-mer Selection

Sept 2024 – Dec 2024

- Implemented a reinforcement learning strategy with specified reward functions to generate variable-length k-mer embeddings from nucleotide-level data dynamically, capturing complex genomic motifs more effectively
- Introduced a novel framework that integrates high-resolution embeddings with flexible, biologically significant sequence vocabularies, enhancing the performance and applicability of gLMs

Predicting mRNA Properties via DNA Foundation LLMs Incorporating Codon Information

March 2024 – Oct 2024

- Developed a universal codon module and designed a mixture-of-expert-based decoder-regressor architecture to integrate codon information into DNA models, allowing the model to prioritize different nucleotides within codons
- Improved a pretrained BERT model by integrating Mamba blocks, accelerating training while maintaining performance
- Achieved superior performance over state-of-the-art models across multiple mRNA downstream tasks, such as SARS-CoV-2 vaccine degradation and mRFP Expression prediction

Diagnosing Alzheimer's Disease via Interpretable Deep Learning Framework

Feb 2024 – Aug 2024

- Developed a quantitative approach combining saliency maps and brain segmentations to assess DL models' focus on brain regions pathologically relevant to Alzheimer's Disease, greatly enhancing interpretability
- Applied two types of gray augmentations on brain MRI data and demonstrated that improved classification accuracy was consistent with enhanced disease focus on critical AD-related regions

Fudan University

Segmenting OCT for Diabetic Macular Edema through Modified U-Net Architecture

Shanghai, China

March 2023 – June 2023

- Designed a novel hybrid loss function by integrating Dice Loss with Cross Entropy Loss, to reduce category imbalances between foreground and background
- Augmented the U-Net architecture by integrating attention gates to accentuate salient features and enhance performance

Predicting Survival Time on Hepatocellular Carcinoma Data with Multi-modal Causal Learning

March 2022 – Feb 2023

- Applied domain-index causal learning to discover causal links between pathological indicator records
- Designed a multi-modality fusion model to predict survival time based on magnetic resonance imaging data and clinical data
- Published a paper on CVPR 2023 as a co-first author, increasing accuracy by 5% via intermediate supervision

Recognizing Oracle Characters through Few-shot Meta-learning Techniques

May 2022 – Aug 2022

- Modified meta-based network by designing more specific decoders in the framework
- Dissected the effect of geometric deformation on few-shot oracle character recognition
- Augmented stroke data extracted from characters and exceeded the SOTA model by 11%

SELECTED ACADEMIC PROJECTS

Carnegie Mellon University

Interpretable Molecular Property Prediction

Pittsburgh, PA, USA

March 2024 - May 2024

- Deployed interpretable Graph Neural Networks (GNNs) for predicting molecular properties by integrating advanced explainers like GNNExplainer and PGExplainer to highlight important molecular substructures
- Designed a functional group masking mechanism to enhance the model’s understanding of molecular structures
- Applied data preprocess on datasets including solubility and biological activity, achieving 2% higher AUC

Prediction of COVID-19 Severity from Whole Blood Biomarkers

March 2024 – May 2024

- Implemented PCA and UMAP for dimensionality reduction, effectively identifying discriminative features from 37,412 RNA-sequencing attributes across 836,148 cell samples to clarify disease mechanisms
- Developed a deep learning model to perform feature-based classification of COVID-19 severity, achieving around 70% accuracy and demonstrating potential applications in precision medicine

Fudan University

Automatic Boundary Segmentation of Intima Interface

Shanghai, China

June 2022 – July 2022

- Utilized Gaussian convolution kernel to construct the edge map, degraded the noise
- Designed a specific cost function for Dual Dynamic Programming to detect the nearly parallel interfaces
- Implemented Dual Snake Model to refine the preliminary segmentation, reduced the standard error

Classification with Lite Vision Transformer on CIFAR-100

March 2022 – June 2022

- Introduced a dual-branch structure to combine LVT and WideResNet to reduce the number of parameters of CrossViT
- Adopted MixToken as a data augmentation method to fit the transformer network
- Improved the performance of LVT on CIFAR-100 by about 2% accuracy

PUBLICATIONS

DNA Language Models for RNA Analyses (arXiv preprint)

Shiyi Du, **Litian Liang**, Jiayi Li, Carl Kingsford*

arXiv:2508.04739, 2025

A Quantitative Approach for Evaluating Disease Focus and Interpretability of Deep Learning Models for Alzheimer’s Disease Classification

Thomas Tam[†], **Litian Liang**[†], Ke Chen, Haohan Wang, Wei Wu*

at the Workshop on Machine Learning and Artificial Intelligence in Bioinformatics and Medical Informatics(MABM), *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2024

Causally-Aware Intraoperative Imputation for Overall Survival Time Prediction

Xiang Li[†], Xuelin Qian[†], **Litian Liang**[†], Lingjie Kong, Qiaole Dong, Jiejun Chen, Dingxia Liu, Xiuzhong Yao, Yanwei Fu* in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023

SKILLS

- **Programming Languages:** Advanced- Python; Intermediate- R, SQL, MATLAB, C++
- **Tool Kits:** PyTorch, Transformers, TensorFlow, Ubuntu
- **Languages:** Fluent- English; Native Speaker- Mandarin

EXTRACURRICULAR ACTIVITIES

- Teaching Assistant for "Computational Molecular Biology and Genomics" at CMU Aug 2024 - Dec 2024
- Teaching Assistant for "Algorithms and Advanced Data Structures" at CMU Jan 2024 - May 2024
- Volunteer at Shanghai Ziqiang Social Service Federation March 2021 – June 2021
- Vice President in the Dralion Union at Fudan University March 2020 – June 2020
- Secretary of the Student Union Sports Department at Fudan University March 2020 – June 2020

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